

## System Requirements & Sizing

These are the minimum and recommended requirements for running the MessageFoundry engine, its message store, and the administration clients. The engine is a headless Python/asyncio service; the console is a separate desktop application.

**On the throughput figures.** MessageFoundry does not publish a validated, guaranteed per-node throughput baseline. The sizing tiers in [Sizing by message volume](#) are **engineering estimates** derived from the architecture and from synthetic load-test profiles — they are starting points, not guarantees. Always establish your own baseline on production-like hardware before go-live (see [Capacity notes](#)).

### Hardware

|              | Minimum (lab / low-volume pilot) | Recommended (single-node production)                                                                                                                                              |
|--------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CPU          | 2 cores                          | 4+ cores (transform throughput is per-core; one worker set per connection)                                                                                                        |
| Memory       | 4 GB                             | 8–16 GB                                                                                                                                                                           |
| Disk         | 10 GB free, any disk             | SSD, 50+ GB or sized to your retention window, on a low-latency local volume                                                                                                      |
| Store volume | —                                | Put the message store on a fast local disk (not a network share). Budget for <b>store + WAL growth</b> : the staged pipeline writes roughly 3× per message on the embedded store. |

The engine and store may share a host for a low-volume pilot. For production, run a **server database** (PostgreSQL or SQL Server) on its own host, sized by your DBA, and keep the engine host dedicated. For volume beyond one CPU core, see [Sizing by message volume](#).

## Operating systems

| Platform                                   | Status                                                                                              |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Windows Server 2022 / 2025</b>          | Primary supported and serviced platform (Windows-service deployment)                                |
| Windows Server 2019                        | Supported                                                                                           |
| Windows 10 / 11                            | Supported (development, pilot, console host)                                                        |
| <b>Linux</b> (modern x86-64 distributions) | Engine supported (cross-platform Python); no bundled service installer — run under systemd yourself |
| macOS                                      | Development / console use only                                                                      |

## Runtime

| Component                 | Requirement                                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Python</b>             | <b>3.11 or later</b> , 64-bit (3.11, 3.12, 3.13 supported; 3.11 and 3.13 are CI-validated)                                                                  |
| Service manager (Windows) | A Windows service wrapper, auto-provisioned and SHA-256-pinned by the installer, or pre-staged. Requires administrator / elevation to register the service. |
| C compiler                | Not required for the default install (runtime dependencies ship as wheels)                                                                                  |

## Databases (message store)

| Database                                | Status                         | Driver / prerequisite                                                                                                                                                                          |
|-----------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SQLite (WAL)</b>                     | Default, bundled — single-node | None ( <code>aiosqlite</code> , in-process)                                                                                                                                                    |
| <b>PostgreSQL 13+</b>                   | Production                     | <code>messagefoundry[postgres]</code> <code>extra ( asyncpg , pure-Python — no OS dependency)</code>                                                                                           |
| <b>Microsoft SQL Server 2019 / 2022</b> | Production                     | <code>messagefoundry[sqlserver]</code> <code>extra ( aiiodbc )</code> <b>plus the OS-level Microsoft ODBC Driver 18 for SQL Server</b> . Read-Committed Snapshot Isolation (RCSI) recommended. |
| MySQL / Oracle                          | Not supported                  | —                                                                                                                                                                                              |

The embedded SQLite store needs no setup and suits pilots and single-node deployments. A **server database is greenfield-only** — there is no in-place migration from a populated SQLite store; drain and cut over. The database tier owns its own backup, HA, and (SQL Server) TDE / purge maintenance. A server database is also the **concurrency / scale substrate** — see below.

## Administration clients

| Client            | Requirement                                                                                                                                                                                                           |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Desktop console   | PySide6 (Qt) desktop application — install with the <code>console</code> extra. Runs on Windows, Linux, or macOS as a separate process; connects to the engine over the HTTP/WebSocket API. <b>Not browser-based.</b> |
| VS Code extension | Visual Studio Code (current stable) — route wizard, validate-on-save, test bench, stage → promote.                                                                                                                    |
| Web browser       | Not required to operate the engine (no web admin UI; the API is HTTP/JSON for tooling).                                                                                                                               |

## Network & ports

| Purpose                          | Default                                          | Notes                                                                                                                                                                                                                         |
|----------------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engine API<br>(HTTP + WebSocket) | 127.0.0.1:8765                                   | <b>Loopback by default</b> , authentication required. To reach the API from another host, front it with a <b>TLS-terminating reverse proxy</b> or tunnel — exposing the listener off loopback should always be done over TLS. |
| Inbound MLLP / TCP listeners     | operator-defined (samples use e.g. 2575 , 2600 ) | Open to sending systems via firewall. Keep MLLP on a <b>trusted network segment</b> .                                                                                                                                         |
| Outbound                         | as configured                                    | Reachability to downstream partners and, for server databases, to the database host.                                                                                                                                          |
| Installer egress                 | HTTPS                                            | Outbound access for the service installer to fetch the pinned service-wrapper binary (or pre-stage it).                                                                                                                       |

**Interface authentication.** Beyond network-segment controls, interfaces can authenticate with **mutual TLS (mTLS)**, **OAuth 2.0 client-credentials**, and **SMART-on-FHIR Backend Services**. Administrative access uses required multi-factor authentication for local accounts; enterprise / Active Directory users authenticate (and get MFA) through their own identity provider via SSO federation.

## Sizing by message volume

**Engineering estimate — not a validated benchmark.** These tiers project the architecture's behavior; they are not committed numbers. Throughput depends heavily on **transform cost per message** (the dominant factor), message size, fan-out, and strict-validation use. **Measure your own feeds** with the load harness before committing (see [Capacity notes](#)).

## How throughput is bounded (read this first)

A single engine process runs **all** message work — decode → peek → route → transform → re-encode — on **one CPU core** (one asyncio event loop; the GIL prevents pure-Python parallelism across threads). So per-process throughput is governed, in order, by:

1. **Transform cost per message** — usually the binding constraint. As an illustrative estimate, a comparable vendor benchmark shows real transformation cutting pass-through throughput by roughly 60% ( $\approx 1000$  msg/s →  $\approx 400$  msg/s). A light / pass-through feed sits near the top of a tier; a heavy transform sits near the bottom.
2. **Durable-write cost** — every stage handoff (ingress → routed → outbound → delivered) is a committed transaction. In-process **SQLite is fastest per write; a server database is slower per single write** (network + MVCC) but is the concurrency substrate (next point).

**To exceed one core**, scale **intra-node** on a server database: many connections / lanes / delivery workers draining **one shared server database** (PostgreSQL or SQL Server) concurrently via `SELECT ... FOR UPDATE SKIP LOCKED` plus row leases. Throughput scales with workers until the **database's commit capacity** is the wall. (SQLite is single-writer and does **not** scale this way — it is the single-process / single-node store.) Engine HA is **single-leader active-passive** — the graph runs on the leader only. A **multi-process, sharded-by-inbound** scale-out (multiple engines, each owning a disjoint set of inbounds, on the shared database) is a **future direction, not a current capability**.

## Tiers

| Tier                         | Peak sustained (est.)            | Indicative daily volume | Deployment shape                                                                       | Store                                                  | Suggested hardware (engine host)                                      |
|------------------------------|----------------------------------|-------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------|
| Pilot / light                | up to ~50 msg/s                  | up to ~1–4 M/day        | 1 process, single node                                                                 | SQLite                                                 | 2 cores / 4 GB                                                        |
| Standard single-node         | ~50–200 msg/s                    | ~4–15 M/day             | 1 process, single node                                                                 | SQLite, or PostgreSQL / SQL Server                     | 4 cores / 8 GB                                                        |
| High single-node             | ~200–500 msg/s                   | ~15–40 M/day            | 1 process, tuned (lean transforms; finite-retry on hot lanes)                          | Server database recommended (PostgreSQL / SQL Server)  | 4–8 cores / 16 GB                                                     |
| High single-node, concurrent | up to ~500 – low-thousands msg/s | up to ~40 M+/day        | 1 process, many connections / lanes draining concurrently via <code>SKIP LOCKED</code> | <b>PostgreSQL / SQL Server</b> (required — not SQLite) | 8+ cores / 32 GB + a dedicated database host sized to the commit load |

## Reading the tiers

- *Peak sustained* is a **per-second** capacity estimate. Healthcare feeds are bursty; real **average** rate (and therefore daily volume) is typically a fraction of peak, so the *indicative daily volume* columns assume a

realistic duty cycle, not `peak × 86,400`.

- The **single-stream / single-core ceiling** is roughly the "High single-node" row — a few hundred msg/s with real transforms, approaching ~1000 msg/s only for light / pass-through work. Past that on one feed you are over one core's budget.
- The **estimated maximum as currently architected** is the "**High single-node, concurrent**" row: one engine process, many connections / lanes draining the shared server database concurrently via `SKIP LOCKED`, bounded by the database's commit ceiling. There is no fixed published cap — on a well-provisioned box with a tuned server database this lands in the **low thousands of msg/s**, beyond which you are **database-bound** and scale the database tier. (A multi-process, sharded-by-inbound scale-out beyond one engine is a **future direction**.) Group-commit and a lazy MSH-only routing peek are identified levers to raise the per-core ceiling.

**Single-stream server-DB caveat.** Because each staged handoff is a committed round-trip, a single delivery worker against a *remote* server database drains far slower than in-process SQLite (a SQL Server CI smoke profile observes roughly 30 deliveries/s for one stream). High volume on a server database comes from **concurrency** — many connections / lanes / processes draining in parallel — not single-stream speed. Size the database host for that concurrent commit load.

### Capacity notes

- Validate with the load harness: run a `smoke` → `fanout-baseline` → `soak` ramp, exercise the `cheap` / `edit` / `slow` transform modes to find your per-core transform ceiling, and compare SQLite vs a server database on identical traffic. Treat the **zero-loss reconciliation** as the headline gate — throughput is meaningless if messages were lost.
- The embedded store has roughly 3× write amplification and a single-writer ceiling; move to PostgreSQL or SQL Server when that becomes the bottleneck.
- Scale **intra-node** on a server database (one delivery worker per outbound; many connections / lanes draining concurrently via `SKIP LOCKED`; keep retry policies finite where head-of-line blocking on a shared FIFO lane would otherwise stall a lane). A multi-process scale-out beyond one engine is a **future direction**. Engine **HA** is **active-passive failover** (opt-in leader / standby cluster on shared PostgreSQL — see [CLUSTERING.md](#)); delegate **database-tier** HA to the database plus a load-balancer VIP.

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